

Millennium-Grade Strategy Upgrade (15 Additions)

This document summarizes and details the 15 additions applied to the intraday post-earnings and backtester. It also documents the operational change to produce fresh predictions every 1 minute from observed history, with no chaining.

Files: • Strategy: bare_strategy_picky.py • Backtester: backtesting_gpu_picky.py

Scope excludes: (2) storage-time truncation to seq_len; (10) dropping 'unknown_date' buckets per user instruction.

Executive Summary: 15 Additions

- Correct SELL semantics: exits now operate in shares, guaranteeing full closures when signal
- Unified trading clock: PREDICT_START, TRADE_START, TRADE_END, LAST_ENTRY = TRADE_END everywhere.
- top_k applied to entries only; exits are never throttled. Exposed via CLI.
- Enforced 20-minute timed exit and -0.50% stop-loss using bid/ask execution realism.
- Spread analytics corrected: mid vs exec recorded per leg; no double-counting.
- True calibration vs trading: predictions logged prior to TRADE_START; orders only within trading window.
- Backtester res CLI arguments (model path, H5 path, etc.) for reproducible runs.
- Deterministic seeding added: order, shuffles, and RNG now reproducible.
- EOD liquidation handled inside the strategy with consistent mechanics.
- Unified trade-log schema (Trade dataclass) used across strategy and backtester.
- Edge > cost hurdle enforced: mu20 must exceed transaction-cost hurdle before
- No magic feature indices; predictions are computed from observed inputs only.
- Caps exposed and enforced: per-name cap and per-minute gross add cap; combined with top_k.
- Tight, data-driven clamp: volatility-scaled clamp ($MAD \rightarrow \sigma$) to bound outliers by intraday regime.
- No prediction chaining: fresh, re-normalized 20-minute forecast computed every minute from observed data

1) SELL semantics (shares vs dollars)

Problem: Exits previously interpreted 'amount' inconsistently, sometimes treating shares as dollars which could leave residual positions.

Change: Implemented explicit sell-by-shares routine and strategy-side calls always pass full position shares.

Effect: Stop-loss, time-based exits, and EOD liquidation now fully close positions as intended.

2) Unified trading clock

Introduced authoritative variables: PREDICT_START, TRADE_START, TRADE_END, LAST_ENTRY pred_len. All gates (logging, entries, exits) and print statements derive from these values; no hard-coded minutes remain.

3) top_k for entries only

At each minute, rank candidates by edge (μ_{20} - hurdle), filter by threshold, then take the first top_k that are not already held. Exits are never limited by top_k.

CLI: --top_k K (None disables ranking cap).

4) Stop-loss and timed exit

Stop-loss triggers when $\text{MTM} \leq -0.50\%$ using bid executions. Timed exit enforces a 20-minute horizon from entry_time, unless stop-loss triggers earlier.

5) Spread analytics corrected

Recorded mid vs exec for each leg; spread_paid computed explicitly as $|exec - mid| * shares$ p
Realized P&L (exec) plus total spread_paid equals hypothetical mid-exec P&L; no double coun

6) Calibration vs trading separation

Predictions are made and logged once $t \geq \max(\text{seq_len}, \text{PREDICT_START})$. Orders only allowed $[\text{TRADE_START}, \text{TRADE_END})$; `LAST_ENTRY` forbids new entries that cannot complete the horizon.

7) Backtester CLI fidelity

Backtester now respects `--model`, `--h5`, `--seq_len`, `--pred_len`, and window parameters; remove hard-codes. Enables reproducibility and batch runs with different configurations.

8) Seeding for determinism

Added a seed routine that sets PYTHONHASHSEED, Python, NumPy, and PyTorch seeds; also enabled cudnn.deterministic. Ensures data ordering, shuffles, and RNG outcomes are reproducible for publication and research sign-off.

9) EOD liquidation inside strategy

Remaining positions are closed within the strategy at TRADE_END (or last bar), guaranteeing consistent execution rules. Backtester no longer performs a separate forecasting regime for E

10) Unified trade-log schema

Introduced a Trade dataclass: (symbol, side, exec_px, mid_px, shares, cash_flow, t, reason, m, spread_paid). Both modules emit trades in this schema, simplifying downstream analytics.

11) Edge > cost hurdle

Computed hurdle = $2 * (\text{spread} + \text{commission} + \text{slippage}) + \text{buffer}$ (in decimal). Entries require $\text{max}(\text{buy_threshold}, \text{hurdle})$. Rank by $(\mu_{20} - \text{hurdle})$.

12) No magic feature indices

All predictions are computed directly from observed windows passed into the model; no hand feature columns. Avoids schema drift and silent data corruption.

13) Caps and gross add limiter

Per-name cap (fraction of initial portfolio) and per-minute gross add cap limit risk-on velocity; combined with top_k to control turnover and capacity usage. CLI: --per_name_cap, --gross_ad

14) Tight, data-driven clamp

Clamps μ_{20} by $k \cdot \sigma_{20m}$ using robust σ from last ~ 60 one-minute returns ($MAD \rightarrow \sigma$), scaling $\sqrt{\text{pred_len}}$. Adapts to regime changes and prevents outlier-driven sizing.

15) No prediction chaining (fresh per-minute forecasts)

Every minute, recompute μ_{20} from the latest observed window; re-normalize per-feature μ_{20} and run a fresh forward pass. No autoregressive feedback of previous predictions into inputs; optional EMA smoothing can be added at the decision layer if desired.